Connecting Probabilistic Databases to Quantum Computing and Tensor Networks via Graphical Models

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Abstract

In the domain of quantum computing for databases, most work has focused on optimizing query processing in relational systems. In this abstract, we extend this view and propose using quantum computers and tensor networks to perform probabilistic inference on probabilistic databases, which is the most expensive operation in probabilistic query evaluation. This theoretical connection shows that quantum computers and tensor networks would play a more profound role in query execution. We discuss and outline a promising research direction that connects probabilistic databases to quantum computers and tensor networks via graphical models. Graphical models have been extensively used in probabilistic databases and quantum computing, yet their role as a bridge between these fields remains unexplored.

Keywords

probabilistic databases, graphical models, tensor networks

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1 Introduction

Probabilistic databases are a generalization of conventional deterministic relational databases [5]. While data has certain values in a deterministic database, in a probabilistic database, the data is modeled using a probability distribution over a so-called possible worlds semantics. The probabilistic view of data is increasingly important today: real life is inherently uncertain, countless applications require probabilistic data [3], and uncertainty is a central feature of modern machine learning and computer science.

One of the key results in the field is the so-called dichotomy theorem [2]. It states that for an important class of queries on probabilistic databases, any such query can be computed either in polynomial time in the database size, or is #P-hard. Given that many queries are computationally intractable, the development of approximations and novel query evaluation methods is well-motivated. Quantum computing has primarily enabled methods for optimizing query processing in relational databases [6]. Connecting quantum computers and tensor networks to support probabilistic inference over probabilistic databases is a natural domain, since both are probabilistic. The unexplored connection among probabilistic databases, quantum computing, and tensor networks is a well-motivated direction for seeking faster probabilistic inference.

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2 Graphical models as a connection

Next, we outline the main theoretical connection between probabilistic databases, quantum computers, and tensor networks. While there are multiple ways to model query evaluation in probabilistic databases [5], a theoretically robust approach is to use so-called probabilistic graphical models [3]. These models enable efficient, large-scale modeling of distributions using probability theory and graphs.

Probabilistic graphical models have been found to have an interesting connection to tensor networks [4]. The key result is that a graphical model on a hypergraph is equivalent to a tensor network on the dual hypergraph. This duality provides a concrete method for translating graphical models arising from probabilistic database instances into tensor networks. The operations in graphical models become tensor contractions.

If one wants to further move the probabilistic inference to a quantum computer, there are methods for rewriting tensor networks using quantum circuits, which are connected to quantum circuit synthesis [1]. The transformation involves embedding the spatial and temporal directions into the tensor network, transforming each tensor into a unitary, and decomposing each unitary into quantum gates. In future research, we will deepen the theoretical connection between these fields, implement the proposed ideas, and experimentally validate whether the suggested methods offer advantages over existing methods for probabilistic inference in probabilistic databases.

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